CYBERSRS

Secure Monitoring Platform Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments. The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes. Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary. The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints

Statement: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business; security; data

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: system collects system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed□□

When: the system initiates collection of system events, authentication activity, process information, and security alerts from multiple departments

Then: the requested inputs are received and represented in the monitoring platform

Method: test

Observable evidence: External test capture of the resulting response

FR-002 - detect unusual login behavior

Statement: The system shall detect unusual login behavior.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system receives a controlled login attempt with an unusual IP address

When: the system detects unusual login behavior

Then: the system identifies the event as anomalous.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-326455fa58e5 is TBD.

FR-003 - detect repeated failed authentication

Statement: The system shall detect repeated failed authentication.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system receives a sequence of failed authentication attempts from the same IP address

When: the system detects repeated failed authentication

Then: the system identifies the event as anomalous.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-66ec7b6022bf is TBD.

FR-004 - detect suspicious process activity

Statement: The system shall detect suspicious process activity.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system observes a process executing with elevated privileges from an unexpected location

When: the system detects suspicious process activity

Then: the system identifies the event as anomalous.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-88a00a650fd2 is TBD.

FR-005 - detect unauthorized configuration changes

Statement: The system shall detect unauthorized configuration changes.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system receives a configuration change request from an unauthorized source

When: the system detects unauthorized configuration changes

Then: the system identifies the event as anomalous.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-51ce9ae15f38 is TBD.

FR-006 - allow Security administrators to review alerts

Statement: The system shall allow Security administrators to review alerts.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: Security administrators review alerts

When: they initiate the review action

Then: the requested alert list is returned in the response

Method: test

Observable evidence: External test capture of the resulting response

FR-007 - allow Security administrators to investigate affected devices

Statement: The system shall allow Security administrators to investigate affected devices.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: Security administrators investigate affected devices

When: they initiate the investigation action

Then: the device status and relevant details are returned in the response

Method: test

Observable evidence: External test capture of the resulting response

FR-008 - allow Security administrators to record investigation notes

Statement: The system shall allow Security administrators to record investigation notes.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: Security administrators record investigation notes

When: they initiate the note recording action

Then: the notes are stored in the system and accessible for review

Method: inspection

Observable evidence: External test capture of the inspection record

FR-009 - allow Security administrators to change alert status

Statement: The system shall allow Security administrators to change alert status.

Rationale: Source obligation E-003_O04.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: Security administrators change alert status

When: they initiate the status update action

Then: the alert status is updated in the system and reflected in the response

Method: test

Observable evidence: External test capture of the resulting response

FR-010 - allow Security administrators to isolate compromised endpoints when necessary

Statement: The system shall allow Security administrators to isolate compromised endpoints when necessary.

Rationale: Source obligation E-003_O05.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: Security administrators isolate compromised endpoints when necessary

When: they initiate the isolation action

Then: the endpoint is placed in a restricted state and no longer participates in network traffic

Method: demonstration

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-3624d3181d5c is TBD.

FR-011 - maintain audit logs of security events and administrator actions

Statement: The system shall maintain audit logs of security events and administrator actions.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system maintains audit logs of security events and administrator actions

When: the system records an event or administrative action

Then: The system maintains audit logs of security events and administrator actions by recording each event or administrative action, with external test capture verifying the resulting response.

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-012 - provide a centralized dashboard for monitoring security status across multiple campus locations

Statement: The system shall provide a centralized dashboard for monitoring security status across multiple campus locations.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system provides a centralized dashboard for monitoring security status across multiple campus locations

When: an administrator accesses the dashboard from any campus location

Then: the dashboard displays security status indicators for all monitored sites

Method: demonstration

Observable evidence: External test capture of the resulting state

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability

When: An analyst inspects or reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

Related requirement ids: FR-001; FR-011; FR-012

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-003; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; NFR-001

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-008; FR-011; FR-012

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

7. Compliance and policy applicability matrix

Open decisions: Decision Q-66cf5513b374 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Security administrators

Detail: Proposed logical endpoint for the source-stated actor: Security administrators.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-006; FR-007; FR-008; FR-009; FR-010

Component name: Actor: Security administrators

Responsibility: Represent the source-stated actor: Security administrators.

Event Collection Module

Detail: Proposed logical owner for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001; NFR-001

Component name: Event Collection Module

Responsibility: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints. The system shall be secure.

Anomaly Detection Engine

Detail: Proposed logical owner for FR-002, FR-003, FR-004, FR-005, NFR-001.

Status: proposed

Evidence ids: E-001; E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; NFR-001

Component name: Anomaly Detection Engine

Responsibility: The system shall detect unusual login behavior. The system shall detect repeated failed authentication. The system shall detect suspicious process activity. The system shall detect unauthorized configuration changes. The system shall be secure.

Security Investigation Workflow

Detail: Proposed logical owner for FR-006, FR-007, FR-008, FR-009, FR-010, NFR-001.

Status: proposed

Evidence ids: E-001; E-003

Requirement ids: FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001

Component name: Security Investigation Workflow

Responsibility: The system shall allow Security administrators to review alerts. The system shall allow Security administrators to investigate affected devices. The system shall allow Security administrators to record investigation notes. The system shall allow Security administrators to change alert status. The system shall allow Security administrators to isolate compromised endpoints when necessary. The system shall be secure.

Audit and Logging Service

Detail: Proposed logical owner for FR-011, NFR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-011; NFR-001

Component name: Audit and Logging Service

Responsibility: The system shall maintain audit logs of security events and administrator actions. The system shall be secure.

Centralized Dashboard

Detail: Proposed logical owner for FR-012, NFR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-012; NFR-001

Component name: Centralized Dashboard

Responsibility: The system shall provide a centralized dashboard for monitoring security status across multiple campus locations. The system shall be secure.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-006, FR-007, FR-008, FR-009, FR-010.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-006; FR-007; FR-008; FR-009; FR-010

From component: Actor: Security administrators

To component: Security Investigation Workflow

Data or action: The system shall allow Security administrators to review alerts. The system shall allow Security administrators to investigate affected devices. The system shall allow Security administrators to record investigation notes. The system shall allow Security administrators to change alert status. The system shall allow Security administrators to isolate compromised endpoints when necessary.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Event Collection Module

To component: Anomaly Detection Engine

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Event Collection Module

To component: Security Investigation Workflow

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Event Collection Module

To component: Audit and Logging Service

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Event Collection Module

To component: Centralized Dashboard

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Anomaly Detection Engine

To component: Event Collection Module

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Anomaly Detection Engine

To component: Security Investigation Workflow

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Anomaly Detection Engine

To component: Audit and Logging Service

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Anomaly Detection Engine

To component: Centralized Dashboard

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Security Investigation Workflow

To component: Event Collection Module

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Security Investigation Workflow

To component: Anomaly Detection Engine

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Security Investigation Workflow

To component: Audit and Logging Service

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Security Investigation Workflow

To component: Centralized Dashboard

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Audit and Logging Service

To component: Event Collection Module

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Audit and Logging Service

To component: Anomaly Detection Engine

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Audit and Logging Service

To component: Security Investigation Workflow

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Audit and Logging Service

To component: Centralized Dashboard

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Centralized Dashboard

To component: Event Collection Module

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Centralized Dashboard

To component: Anomaly Detection Engine

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Centralized Dashboard

To component: Security Investigation Workflow

Data or action: The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: NFR-001

From component: Centralized Dashboard

To component: Audit and Logging Service

Data or action: The system shall be secure.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-006, FR-007, FR-008, FR-009, FR-010.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-006; FR-007; FR-008; FR-009; FR-010

Untrusted component: Actor: Security administrators

Trusted component: Security Investigation Workflow

Enforcement: The system shall allow Security administrators to review alerts. The system shall allow Security administrators to investigate affected devices. The system shall allow Security administrators to record investigation notes. The system shall allow Security administrators to change alert status. The system shall allow Security administrators to isolate compromised endpoints when necessary.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Security administrators

Goal: The system shall allow Security administrators to review alerts.

Story: As Security administrators, I need the following behavior: The system shall allow Security administrators to review alerts.

Preconditions

1. Security administrators review alerts

Steps

1. they initiate the review action

Success outcome: the requested alert list is returned in the response

Requirement ids: FR-006

Evidence ids: E-003

Id: UC-002

Actor: Security administrators

Goal: The system shall allow Security administrators to investigate affected devices.

Story: As Security administrators, I need the following behavior: The system shall allow Security administrators to investigate affected devices.

Preconditions

1. Security administrators investigate affected devices

Steps

1. they initiate the investigation action

Success outcome: the device status and relevant details are returned in the response

Requirement ids: FR-007

Evidence ids: E-003

Id: UC-003

Actor: Security administrators

Goal: The system shall allow Security administrators to record investigation notes.

Story: As Security administrators, I need the following behavior: The system shall allow Security administrators to record investigation notes.

Preconditions

1. Security administrators record investigation notes

Steps

1. they initiate the note recording action

Success outcome: the notes are stored in the system and accessible for review

Requirement ids: FR-008

Evidence ids: E-003

Id: UC-004

Actor: Security administrators

Goal: The system shall allow Security administrators to change alert status.

Story: As Security administrators, I need the following behavior: The system shall allow Security administrators to change alert status.

Preconditions

1. Security administrators change alert status

Steps

1. they initiate the status update action

Success outcome: the alert status is updated in the system and reflected in the response

Requirement ids: FR-009

Evidence ids: E-003

Id: UC-005

Actor: Security administrators

Goal: The system shall allow Security administrators to isolate compromised endpoints when necessary.

Story: As Security administrators, I need the following behavior: The system shall allow Security administrators to isolate compromised endpoints when necessary.

Preconditions

1. Security administrators isolate compromised endpoints when necessary

Steps

1. they initiate the isolation action

Success outcome: the endpoint is placed in a restricted state and no longer participates in network traffic

Requirement ids: FR-010

Evidence ids: E-003

Unresolved decisions: Decision Q-3624d3181d5c is TBD.

Open issues: Decision Q-3624d3181d5c is TBD.

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: system collects system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed□□
2. When: the system initiates collection of system events, authentication activity, process information, and security alerts from multiple departments

Expected result: Expected: the requested inputs are received and represented in the monitoring platform

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: the system receives a controlled login attempt with an unusual IP address
2. When: the system detects unusual login behavior

Expected result: Expected: the system identifies the event as anomalous.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: the system receives a sequence of failed authentication attempts from the same IP address
2. When: the system detects repeated failed authentication

Expected result: Expected: the system identifies the event as anomalous.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: the system observes a process executing with elevated privileges from an unexpected location
2. When: the system detects suspicious process activity

Expected result: Expected: the system identifies the event as anomalous.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: the system receives a configuration change request from an unauthorized source
2. When: the system detects unauthorized configuration changes

Expected result: Expected: the system identifies the event as anomalous.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: Security administrators review alerts
2. When: they initiate the review action

Expected result: Expected: the requested alert list is returned in the response

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: Security administrators investigate affected devices
2. When: they initiate the investigation action

Expected result: Expected: the device status and relevant details are returned in the response

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: Security administrators record investigation notes
2. When: they initiate the note recording action

Expected result: Expected: the notes are stored in the system and accessible for review

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: Security administrators change alert status
2. When: they initiate the status update action

Expected result: Expected: the alert status is updated in the system and reflected in the response

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: Security administrators isolate compromised endpoints when necessary
2. When: they initiate the isolation action

Expected result: Expected: the endpoint is placed in a restricted state and no longer participates in network traffic

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: system maintains audit logs of security events and administrator actions

2. When: the system records an event or administrative action

Expected result: Expected: The system maintains audit logs of security events and administrator actions by recording each event or administrative action, with external test capture verifying the resulting response.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-012

Requirement ids: FR-012

Procedure

1. Given: system provides a centralized dashboard for monitoring security status across multiple campus locations
2. When: an administrator accesses the dashboard from any campus location

Expected result: Expected: the dashboard displays security status indicators for all monitored sites

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-013

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability
2. When: An analyst inspects or reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to favor one SDLC model over others. All models receive equal suitability scores (40/100) because the decision factors are missing or unranked. The absence of explicit requirements about budget, delivery, failure consequences, feedback, legacy systems, regulation, schedule, stability, uncertainty, verification, or assurance levels prevents meaningful differentiation between lifecycle approaches.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.; FR-002: The system shall detect unusual login behavior.; FR-003: The system shall detect repeated failed authentication.; FR-004: The system shall detect suspicious process activity.; FR-005: The system shall detect unauthorized configuration changes.; FR-006: The system shall allow Security administrators to review alerts.; FR-007: The system shall allow Security administrators to investigate affected devices.; FR-008: The system shall allow Security administrators to record investigation notes.; FR-009: The system shall allow Security administrators to change alert status.; FR-010: The system shall allow Security administrators to isolate compromised endpoints when necessary.; FR-011: The system shall maintain audit logs of security events and administrator actions.; FR-012: The system shall provide a centralized dashboard for monitoring security status across multiple campus locations.; NFR-001: The system shall be secure.
2. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.
2. Review implementation and acceptance evidence against FR-003: The system shall detect repeated failed authentication.
3. Review implementation and acceptance evidence against FR-006: The system shall allow Security administrators to review alerts.
4. Review implementation and acceptance evidence against FR-009: The system shall allow Security administrators to change alert status.
5. Review implementation and acceptance evidence against FR-010: The system shall allow Security administrators to isolate compromised endpoints when necessary.
6. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.; FR-002: The system shall detect unusual login behavior.; FR-003: The system shall detect repeated failed authentication.; FR-004: The system shall detect suspicious process activity.; FR-005: The system shall detect unauthorized configuration changes.; FR-006: The system shall allow Security administrators to review alerts.; FR-007: The system shall allow Security administrators to investigate affected devices.; FR-008: The system shall allow Security administrators to record investigation notes.; FR-009: The system shall allow Security administrators to change alert status.; FR-010: The system shall allow Security administrators to isolate compromised endpoints when necessary.; FR-011: The system shall maintain audit logs of security events and administrator actions.; FR-012: The system shall provide a centralized dashboard for monitoring security status across multiple campus locations.; NFR-001: The system shall be secure.

3. In an authorized isolated test environment, execute TEST-001: Given: system collects system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed□□ When: the system initiates collection of system events, authentication activity, process information, and security alerts from multiple departments Expected: the requested inputs are received and represented in the monitoring platform
4. In an authorized isolated test environment, execute TEST-002: Given: the system receives a controlled login attempt with an unusual IP address When: the system detects unusual login behavior Expected: the system identifies the event as anomalous.
5. In an authorized isolated test environment, execute TEST-003: Given: the system receives a sequence of failed authentication attempts from the same IP address When: the system detects repeated failed authentication Expected: the system identifies the event as anomalous.
6. In an authorized isolated test environment, execute TEST-004: Given: the system observes a process executing with elevated privileges from an unexpected location When: the system detects suspicious process activity Expected: the system identifies the event as anomalous.
7. In an authorized isolated test environment, execute TEST-005: Given: the system receives a configuration change request from an unauthorized source When: the system detects unauthorized configuration changes Expected: the system identifies the event as anomalous.
8. In an authorized isolated test environment, execute TEST-006: Given: Security administrators review alerts When: they initiate the review action Expected: the requested alert list is returned in the response
9. In an authorized isolated test environment, execute TEST-007: Given: Security administrators investigate affected devices When: they initiate the investigation action Expected: the device status and relevant details are returned in the response
10. In an authorized isolated test environment, execute TEST-008: Given: Security administrators record investigation notes When: they initiate the note recording action Expected: the notes are stored in the system and accessible for review
11. In an authorized isolated test environment, execute TEST-009: Given: Security administrators change alert status When: they initiate the status update action Expected: the alert status is updated in the system and reflected in the response
12. In an authorized isolated test environment, execute TEST-010: Given: Security administrators isolate compromised endpoints when necessary When: they initiate the isolation action Expected: the endpoint is placed in a restricted state and no longer participates in network traffic
13. In an authorized isolated test environment, execute TEST-011: Given: system maintains audit logs of security events and administrator actions When: the system records an event or administrative action Expected: The system maintains audit logs of security events and administrator actions by recording each event or administrative action, with external test capture verifying the resulting response.
14. In an authorized isolated test environment, execute TEST-012: Given: system provides a centralized dashboard for monitoring security status across multiple campus locations When: an administrator accesses the dashboard from any campus location Expected: the dashboard displays security status indicators for all monitored sites
15. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability When: An analyst inspects or reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
16. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
17. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.
2. Review implementation and acceptance evidence against FR-003: The system shall detect repeated failed authentication.
3. Review implementation and acceptance evidence against FR-006: The system shall allow Security administrators to review alerts.
4. Review implementation and acceptance evidence against FR-009: The system shall allow Security administrators to change alert status.
5. Review implementation and acceptance evidence against FR-010: The system shall allow Security administrators to isolate compromised endpoints when necessary.
6. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments for university servers and managed endpoints.
2. Review implementation and acceptance evidence against FR-003: The system shall detect repeated failed authentication.
3. Review implementation and acceptance evidence against FR-006: The system shall allow Security administrators to review alerts.
4. Review implementation and acceptance evidence against FR-009: The system shall allow Security administrators to change alert status.
5. Review implementation and acceptance evidence against FR-010: The system shall allow Security administrators to isolate compromised endpoints when necessary.
6. Review implementation and acceptance evidence against NFR-001: The system shall be secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Id: E-002

Source: description

Quote: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Id: E-003

Source: description

Quote: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Id: E-004

Source: description

Quote: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7468

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7429

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7322

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7183

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.7274

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:19 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 19

Page or section: text chunk 20

Relevance score: 0.7272

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.32

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-002	TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-004	E-002	TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-005	E-002	TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-006	E-003	UC-001, TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-007	E-003	UC-002, TEST-007, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-008	E-003	UC-003, TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-009	E-003	UC-004, TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-010	E-003	UC-005, TEST-010, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-011	E-004	TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-012	E-004	TEST-012, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-001	TEST-013, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.32